

Heterogeneous Agent Trade

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Introduction

- ▶ **Classic Question:** What are the gains from trade? In representative agent models:

→ Arkolakis et al. (2012): $\text{gains} = \underbrace{\text{elasticity}}_{\frac{d \log \text{Imports}_{i,j}}{d \log d_{i,j}}} \cdot \Delta \text{Own-Consumption}$

→ Applicable to a large class of models: Ricardian, Armington,...

- ▶ This result assumes trade shocks effect agents uniformly
- ▶ Recent evidence points to heterogeneity across expenditure:

Auer et al. (2022), Borusyak and Jaravel (2021), Colicev et al. (2022)

This paper:

- ▶ HA + Incomplete-Market + multi-country trade
- ▶ Study implications for gains from trade

Model:

- ▶ Armington-style trade with heterogeneous household
 - Countries produce a single differentiated goods
 - Discrete choice over world varieties
 - HHs face uninsurable idiosyncratic risk
 - Access to r.f. debt limited by borrowing constraint

Results:

- ▶ Elasticity of trade higher for poor households
- ▶ Gains from trade are higher for poor households
- ▶ Average gains are 3-times larger relative to the ACR estimates

Model: Environment

- ▶ M countries, each producing one differentiated goods
- ▶ Representative firm in country i produces $Y_i = A_i N_i$
 - N_i is Labor, A_i is TFP
 - Pay wages w_i to workers (local, no migration!)
 - Trade subject to iceberg costs $d_{i,j} \geq 1$ with $d_{i,i} = 1$ between source j and destination i
 - Perfectly competitive: $p_{i,j} = d_{i,j} \frac{w_j}{A_j}$
- ▶ Mass L_i of workers in each country $i \in \{1, \dots, M\}$

Model: Households

- ▶ Immobile with discounted utility: $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \tilde{v}(\{c_{i,j,t}\}_M)$

$$\tilde{v}(c_{i,j,t}) = \sum_{j=1}^M \iota_{i,j,t} \tilde{u}(c_{i,j,t})$$

$$\tilde{u}(c_{i,j,t}) = u(c_{i,j,t}) + \varepsilon_{i,j,t}$$

$$\varepsilon_{i,j,t} \sim F(\varepsilon) = \exp \left(- \exp \left(- \frac{\varepsilon}{\sigma_\varepsilon} \right) \right)$$

- ▶ Can save/borrow in non-contingent pure-exchange asset $a \geq -\phi_i$
- ▶ Labor productivity $z'|z \sim \mathcal{P}(z'|z)$
- ▶ Conditional on sourcing from j , budget constraint is

$$p_{i,j} c_{i,j,t} + a_{t+1} \leq R_i a_t + w_i z$$

Model: Households

- Value of household in country i in state (a, z)

$$\begin{aligned}v_i(a, z) &= \mathbb{E}_\varepsilon \max_j \{v_i(a, z, j) + \varepsilon_j\} \\&= \sigma_\varepsilon \log \left\{ \sum_j \exp \left(\frac{v_i(a, z, j)}{\sigma_\varepsilon} \right) \right\} \\&= \sigma_\varepsilon \log \Phi_i(a, z)\end{aligned}$$

- Value before ε_j are realized

$$\begin{aligned}v_i(a, z, j) &= \max_{a'} u(c_{i,j}) + \beta \mathbb{E}[v_i(a', z')] \\ \text{s.t.} \quad & p_{i,j} c_{i,j} + a' \leq R_i a + w_i z\end{aligned}$$

- Define sourcing probabilities

$$\begin{aligned}\pi_{i,j}(a, z) &\equiv \text{Prob}(j = \arg\max_k v_i(a, z, k) + \varepsilon_k) \\&= \frac{\exp \left(\frac{v_i(a, z, j)}{\sigma_\varepsilon} \right)}{\Phi_i(a, z)}\end{aligned}$$

Model: Euler Equation

- Go back to the value function

$$\begin{aligned}v_i(a, z) &= \sigma_\varepsilon \log \Phi_i(a, z) \\&= \sigma_\varepsilon \log \left(\frac{\exp(\frac{v_i(a, z, i)}{\sigma_\varepsilon})}{\pi_{i,i}(a, z)} \right) \\v_i(a, z) &= -\sigma_\varepsilon \log \pi_{i,i}(a, z) + v_i(a, z, i)\end{aligned}$$

→ Ex-ante value can be expressed as home choice probability + home value of consumption

→ $\pi_{i,i}$ summarizes expected value of taste shocks

- Consequence for E.E

$$\frac{u'(c_i(a, z, j))}{p_{i,j}} = \beta \mathbb{E} \left\{ \frac{u'(c_i(a, z, i))R_i}{p_{i,i}} - \frac{\sigma_\varepsilon}{\pi_{i,i}(a, z)} \frac{\partial \pi_{i,i}(a, z)}{\partial a} \right\}$$

→ Equate MU with changes in variety and returns to savings in "home utils" tomorrow

Model: Aggregation

- Distribution of households in country i

$$\lambda_i(a', z') = \sum_j \int \int \pi_{i,j}(a, z) \mathcal{P}(z'|z) \lambda_i(a', z') \mathbb{I}(g_i(a, z, j) = a') da dz$$

- Bilateral trade

$$M_{i,j} = L_i \int \int p_{i,j} c_i(a, z, j) \pi_{i,j}(a, z) \lambda_i(a, z) da dz$$

$$X_{i,j} = L_j \int \int p_{j,i} c_j(a, z, i) \pi_{j,i}(a, z) \lambda_j(a, z) da dz$$

$$\widetilde{PC}_i = L_i \int \int p_{i,j} c_i(a, z, j) \pi_{i,j}(a, z) \lambda_i(a, z) da dz$$

- Total assets: $\mathcal{A}'_i = L_i \sum_j g_i(a, z, j) \pi_{i,j}(a, z) \lambda_i(a, z) da dz$
- Labor: $N_i = L_i \int \int z \lambda_i(a, z) da dz$
- Output:

$$\begin{aligned} p_{i,i} Y_i &= w_i N_i \\ &= p_{i,i} \sum_j L_j \int \int d_{j,i} c_j(a, z, i) \pi_{j,i}(a, z) \lambda_j(a, z) da dz \end{aligned}$$

Model: Aggregation

- ▶ Using aggregation and integrating over budget constraint

$$p_{i,i} Y_i = \widetilde{P} C_i + \sum_{j \neq i} X_{j,i} - \sum_{j \neq i} M_{i,j}$$

$$p_{i,i} Y_i = \widetilde{P} C_i + \mathcal{A}'_i - R_i \mathcal{A}_i$$

Equilibrium: Policies $\{g_i(a, z, j)\}$, distribution $\lambda_i(a, z)$, and prices $\{w_i, p_{i,j}, R_i\}_{i,j}$ such that

- ▶ Given $\{w_i, p_{i,j}, R_i\}_{i,j}$, $g_i(a, z, j)$ is the optimal policy for the household problem
- ▶ Good's Market Clears $p_{i,i} Y_i = \sum_j X_{j,i}$
- ▶ Asset market clears: **Two options**
 1. Integrated: $\sum_j \mathcal{A}'_j = 0 \Rightarrow R_i = R$
 2. Autarkic: $\mathcal{A}'_j = 0 \Rightarrow$ separate rate R_i for every country

- ▶ Characterize two important and related objects in the model:

1. Trade Elasticity: $\theta_{i,j} \equiv \frac{\partial \log \frac{M_{i,j}}{M_{i,i}}}{\partial \log d_{i,j}}$

2. Welfare Effect of Trade-shocks: $\frac{dv_i(a,z)}{d \log d_{i,j}}$

- ▶ Consider special log-utility case
- ▶ Quantify the gains from trade for poor and rich

- How do imports in country i of goods j respond to changes in trade-cost $d_{i,j}$?

$$\theta_{i,j} = \frac{d_{i,j}}{M_{i,j}} \frac{\partial M_{i,j}}{\partial d_{i,j}} - \frac{d_{i,j}}{M_{i,i}} \frac{\partial M_{i,i}}{\partial d_{i,j}}$$

$$M_{i,j} = L_i \int \int p_{i,j} c_i(a, z, j) \pi_{i,j}(a, z) \lambda_i(a, z) da dz$$

- ▶ Trade Elasticity between country i and j is

$$\theta_{i,j} = 1 + \int_{z,a} \left\{ \theta'_{i,j}(a, z) + \theta^E_{i,j}(a, z) \right\} \omega_{i,j}(a, z) da dz - \int_{z,a} \left\{ \theta'_{ii,j}(a, z) + \theta^E_{ii,j}(a, z) \right\} \omega_{i,i}(a, z) da dz$$

- ▶ Micro-level elasticities

$$\theta'_{i,j} = \frac{\partial \log c_i(a, z, j)}{\partial \log d_{i,j}}$$

$$\theta^E_{i,j} = \frac{\partial \log \pi_{i,j}(a, z)}{\partial \log d_{i,j}}$$

$$\theta'_{ii,j} = \frac{\partial \log c_i(a, z, i)}{\partial \log d_{i,j}}$$

$$\theta^E_{ii,j} = \frac{\partial \log \pi_{i,i}(a, z)}{\partial \log d_{i,j}}$$

- From budget constraint...

$$\underbrace{\frac{\partial \log c_i(a, z, j)}{\partial \log d_{i,j}}}_{\theta_{i,j}^l(a, z)} = - \underbrace{\frac{\partial \log p_{i,j}}{\partial \log d_{i,j}}}_{\text{pass-through}} \left[\frac{p_{i,j}}{p_{i,j} c_i(a, z, j)} \frac{\partial g_i(a, z, j)}{\partial p_{i,j}} + 1 \right]$$

→ If household is constrained, $\theta_{i,j}^l(a, z) = -1$

→ If household is unconstrained, $\theta_{i,j}^l(a, z) < -1$

→ **Poor household more price sensitive**

Trade Elasticity: Extensive Margin

- ▶ From definition of $\pi_{i,j}(a, z)$...

$$\underbrace{\frac{\partial \log \pi_{i,j}(a, z)}{\partial \log d_{i,j}}}_{\theta_{i,j}^E(a, z)} = \frac{1}{\sigma_e} \frac{\partial v_i(a, z, j)}{\partial \log d_{i,j}} - \frac{d_{i,j}}{\Phi_i(a, z)} \frac{\partial \Phi_i(a, z)}{\partial d_{i,j}}$$

- ▶ Second term analogous to changes in price-index in CES model
- ▶ Take M to be very large, therefore $\frac{\partial \Phi_i(a, z)}{\partial d_{i,j}} \approx 0$ and $\frac{\partial v_i(a, z)}{\partial d_{i,j}} \approx 0$

$$\theta_{i,j}^E(a, z) \approx -\frac{1}{\sigma_e} u'(c_i(a, z, j)) c_i(a, z, j)$$

- ▶ Changes in prob. of consuming variety j depends on the elasticity of utility
- ▶ Substitution to j is high if lots of utility can be derived from it's consumption
- ▶ Question: How does this margin depend on wealth? What's $\frac{\partial \theta_{i,j}^E(a, z)}{\partial a}$?

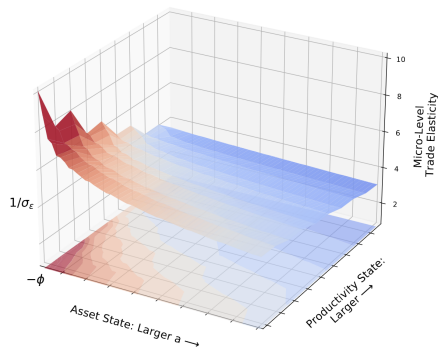
Trade Elasticity: Extensive Margin (Cont.)

$$\frac{\partial \theta_{i,j}^E(a, z)}{\partial a} = \underbrace{\frac{\partial c_i(a, z, j)}{\partial a}}_{\text{MPC!}} u'(c_i(a, z, j)) \left[1 + \underbrace{\frac{u''(c_i(a, z, j)) c_i(a, z, j)}{u'(c_i(a, z, j))}}_{\text{Risk Aversion}} \right]$$
$$\frac{\partial \theta_{i,j}^E(a, z)}{\partial a} = \text{MPC}_i(a, z, j) u'(c_i(a, z, j)) [1 - \gamma_i(a, z, j)]$$

- ▶ Extensive margin varies across wealth depending on risk aversion (γ) and MPC
- ▶ CRRA Case: $\gamma_i(a, z, j) = \gamma$
 - If $\gamma > 1$, high MPC households have a strong substitution effect
 - If $\gamma = 1$, $\theta_{i,j}^E(a, z) = -\frac{1}{\sigma_e}$

Trade Elasticity: Visualization

Household-Level Elasticities $-\theta_{i,j}(a, z)$



- ▶ Poor households near $-\phi$ sensitive to trade shocks
- ▶ Rich households have elasticity equal to $-\frac{1}{\sigma_\epsilon}$
- ▶ In complete-markets: $\theta_{i,j} = -\frac{1}{\sigma_\epsilon}$ for all agents

- ▶ How does welfare change across steady states after trade costs fall?

$$v_i(a, z) = -\sigma_e \log \pi_{i,i}(a, z) + u(c_i(a, z, i)) + \beta \mathbb{E}[v_i(g_i(a, z, i), z')]$$

Gains from Trade

- How does welfare change across steady states after trade costs fall?

$$\begin{aligned} \frac{dv_i(a, z)}{d \log d_{i,j}} = & -\sigma_e \frac{d \log \pi_{i,i}}{d \log d_{i,j}} && \text{Gains from Substitution} \\ & + u'(c_i(a, z, i)) a \frac{d \frac{R}{p_{i,i}}}{d \log d_{i,j}} && \text{Gains from Factor Prices} \\ & + \left[-\frac{u'(c_i(a, z, i))}{p_{i,j}} + \beta \mathbb{E} \left[\frac{\partial v_i(a, z)}{\partial z} \right] \right] \frac{dg_i(a, z, i)}{d \log d_{i,j}} && \text{Gains from Savings} \\ & + \beta \mathbb{E}[\dots \text{repeat} \dots] \end{aligned}$$

- How does welfare change across steady states after trade costs fall?

$$\frac{dv_i(a, z)}{d \log d_{i,j}} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \{A(a_t, z_t) + B(a_t, z_t) + C(a_t, z_t)\}$$

$$A(a_t, z_t) = -\sigma_e \frac{d \log \pi_{i,i}(a, z)}{d \log d_{i,j}}$$

$$B(a_t, z_t) = u'(c_i(a, z, i)) a \frac{d \frac{R}{p_{i,i}}}{d \log d_{i,j}}$$

$$C(a_t, z_t) = \left[-\frac{u'(c_i(a, z, i))}{p_{i,j}} + \beta \mathbb{E} \left[\frac{\partial v_i(a, z)}{\partial z} \right] \right] \frac{dg_i(a, z, i)}{d \log d_{i,j}}$$

Gains from Substitution

$$A(a, z) = -\sigma_\varepsilon \frac{d \log \pi_{i,i}(a, z)}{d \log d_{i,j}}$$

- Reflects the substitution effect between varieties in response to changes in trade costs

$$\frac{d \log \pi_{ii}(a, z)}{d \log d_{i,j}} = \sum_k \pi_{i,k}(a, k) \left[\bar{\theta}_{ii,j}^E(a, z) - \bar{\theta}_{ik,j}^E(a, z) \right]$$

- If cross partials $\bar{\theta}_{ik,j}^E = \frac{d \log \pi_{i,k}(a, z)}{d \log d_{i,j}} \approx 0$ for all $k \neq j$, then

$$A(a, z) \approx \sigma_\varepsilon \cdot \pi_{i,j}(a, z) \cdot \bar{\theta}_{ij,j}^E(a, z)$$

- Gains from substitution depend on ...

1. Taste-shock parameter (σ_ε)
2. Initial Exposure: ($\pi_{i,j}$)
3. Elasticity of trade: $\bar{\theta}_{i,j}^E$

Gains from Trade: Log Utility

1. Trade elasticity is only a function of taste

$$\theta = -\frac{1}{\sigma_\varepsilon}$$

2. Standard Gravity equation

$$\frac{M_{i,j}}{M_{i,i}} = \frac{\pi_{i,j}}{\pi_{i,i}} = \left(\frac{w_j/A_j}{w_i/A_i} \right)^{-\frac{1}{\sigma_\varepsilon}} d_{i,j}^{-\frac{1}{\sigma_\varepsilon}}$$

3. Gains from substitution are homogenous across agents

$$\frac{dv_i(a, z)}{d \log d_{i,j}} = \frac{1}{\theta(1 - \beta)} \frac{d \log \pi_{i,i}}{d \log d_{i,j}} + \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \{ B(a_t, z_t) + C(a_t, z_t) \}$$

Implication:

- ▶ Heterogeneity doesn't matter for trade elasticity
- ▶ Trade flows are not related to household heterogeneity
- ▶ Substitution effect looks like the ACR "Gains from Trade"

Gains from Trade: Log Utility (Cont.)

Why the separation between household heterogeneity and trade with log utility?

- Consider the household's problem

$$\begin{aligned} v_i(a, z, j) &= \max_{a' > -\phi_i} \left\{ \log \left(\frac{Ra + w_i z - a'}{p_{i,j}} \right) + \beta \mathbb{E}[v_i(a', z')] \right\} \\ &= \max_{a' > -\phi_i} \left\{ \log (Ra + w_i z - a') + \beta \mathbb{E}[v_i(a', z')] \right\} - \log p_{i,j} \end{aligned}$$

→ a' independent of $p_{i,j}$ thus $\theta'_{i,j}(a, z) = -1 \Rightarrow$ **Crucial for constant trade elasticity**

→ expenditures are equalized: $p_{i,i}c_i(a, z, j) = p_{i,j}c_i(a, z, j) \Rightarrow$ **Crucial for Gravity Equation**

- Lastly, because of the log-linearity of the value w.r.t price of goods

$$\pi_{i,j}(a, z) = \frac{\exp \left(\frac{v_i(a, z, j)}{\sigma_\varepsilon} \right)}{\sum_k \exp \left(\frac{v_i(a, z, k)}{\sigma_\varepsilon} \right)} = \frac{\exp \left(\frac{-\log p_{i,j}}{\sigma_\varepsilon} \right)}{\sum_k \exp \left(\frac{-\log p_{i,k}}{\sigma_\varepsilon} \right)} \Rightarrow \frac{1}{\theta} \frac{d \log \pi_{i,i}}{d \log d_{i,j}} = \sum_k \pi_{i,k} \left\{ \frac{d \log p_{i,k}}{d \log d_{i,j}} - \frac{d \log p_{i,i}}{d \log d_{i,j}} \right\}$$

Digression: What about Eaton-Kortum?

- ▶ Armington is tractable but unappealing, somewhat easy to redo in Ricardian fashion
- ▶ Some differences:
 - ▶ Continuum of goods; $i \in [0, 1]$, each consumer wants a CES aggregate with elasticity σ
 - ▶ A country i , produces $i \in [0, 1]$ with productivity $z_i(m)$ and $z_i(m) \sim F(z) = \exp(-T_i z^{-\theta})$
 - ▶ In country i , price of good m is $p_i(m) = \min_j \{P_j(m)\}$ where $P_j(m) = \frac{w_j}{z_j(m)} d_{i,j}$
- ▶ Standard objects: $\pi_{i,j} = \text{Prob}(j = \text{argmin}_k \{P_k(m)\}) = \frac{T_j (d_{i,j} w_j)^{-\theta}}{\Phi_i}$
- ▶ One big difference: Households substitute identically across m goods, source identically

$$\theta'_{i,j}(a, z) = - \left[\frac{d_{i,j}}{P_i c} \frac{\partial a'}{\partial d_{i,j}} + \pi_{i,j} \right]$$

$$\frac{\partial \log M_{i,j}}{\partial \log d_{i,j}} = \frac{\partial \log \pi_{i,j}}{\partial \log d_{i,j}} + \int_{z,a} \theta'_{i,j}(a, z) \omega_{i,j}(a, z) da dz$$

→ Intensive margin is the source of heterogeneity in the trade elasticity

Calibration

Two modifications:

- ▶ Add home-bias: $u(c_{i,j}) + \psi_j(z) + \varepsilon_j$ where $\psi_i(z) = \bar{\psi}z$ and zero otherwise
 - ▶ Such that import shares are the same for poor and wealthy h.h. as in Borusyak and Jaravel (2021)
- ▶ Country specific taste shocks and borrowing constraint
 - ▶ $\sigma_{\varepsilon,i} = \sigma_{\varepsilon} A_i^{1-\gamma} \Rightarrow$ TFP differences don't contribute to trade elasticity differences
 - ▶ ϕ_i is 50-percent autarky labor income $\Rightarrow$ HH debt level is invariant to autarky income

Strategy

- ▶ Use Eaton and Kortum (2002) trade flow data for 19 countries
- ▶ Use to identify A_i and $d_{i,j}$
- ▶ Reduce dimensions of iceberg costs

$$\log d_{i,j} = \underbrace{d_k}_{\text{distance}} + \underbrace{b}_{\text{border}} + \underbrace{\ell}_{\text{language}} + \underbrace{e_h}_{\text{euros}} + \underbrace{m_i}_{\text{importer}}$$

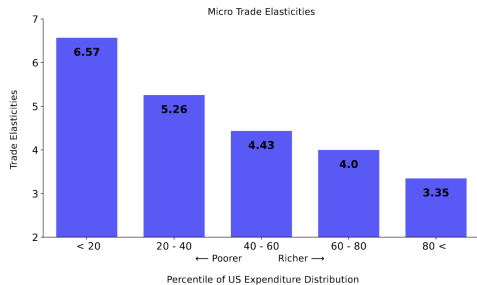
- ▶ No gravity equation if $\gamma > 1$, estimate following in model and data and match coefficients

$$\log \left(\frac{M_{i,j}}{M_{i,i}} \right) = \text{im}_i + \text{ex}_j + d_k + b + \ell + e_h + \delta_{i,j}$$

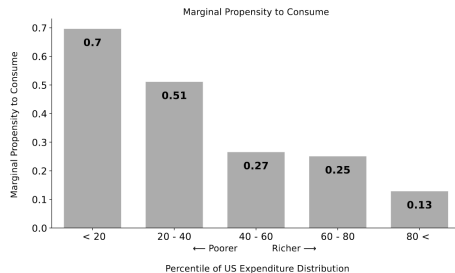
Description	Value	Target
Discount Factor, β	0.92	Global Interest Rate of 1%
CRRA parameter, γ	1.45	} Micro elasticities of Auer et al. (2022)
Type One E-V parameter, $1/\sigma_\epsilon$	3.0	
Slope of Quality Shifter, $\psi_{ii}(z)$	0.72	Micro moments of Borusyak and Jaravel (2021)
Borrowing Constraint ϕ_i	—	50% of i 's autarky labor income
Income Process on z	—	Krueger, Mitman, and Perri (2016)

Trade Elasticity and MPC

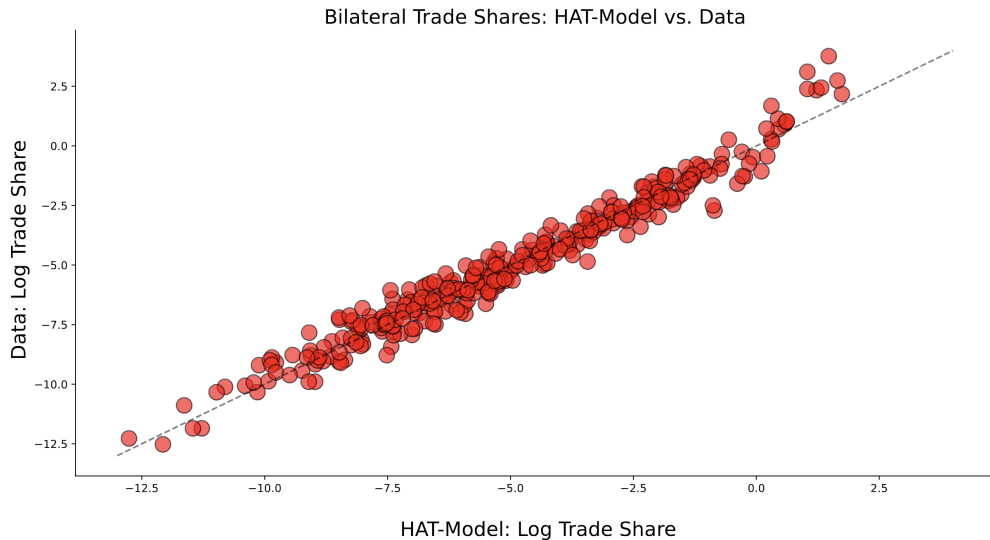
Household-Level Elasticities $-\theta_{i,j}(a, z)$

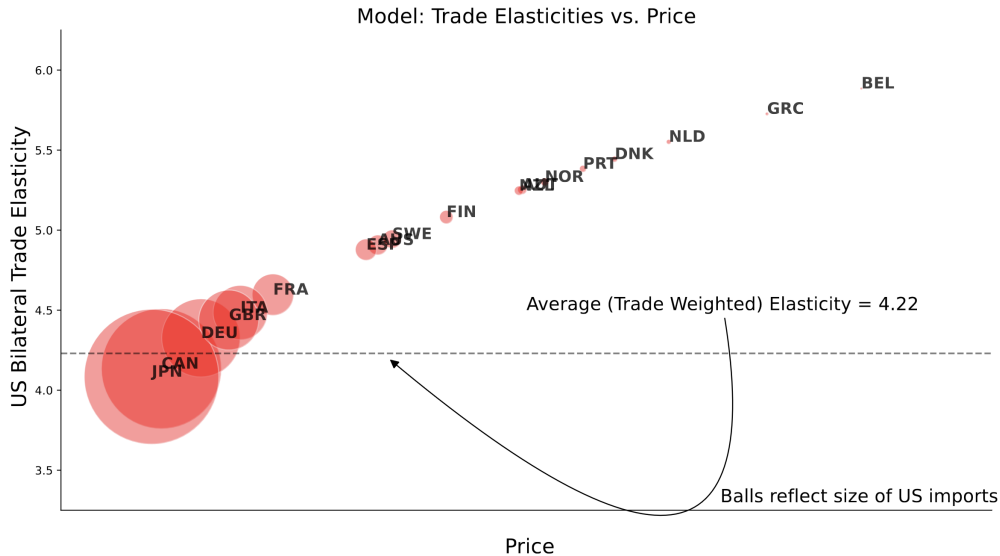


Household-Level MPCs



Bilateral Trade



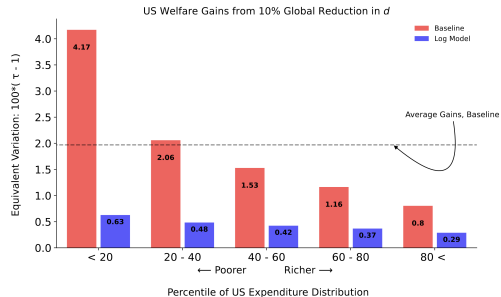


- ▶ Two experiments: (1) Unilateral ($d_{US,j}$ falls for all j); (2) Globalization ($d_{i,j}$ falls for all (i,j))
- ▶ Welfare measured along transition in terms of equivalent variation in period wealth

$$p_{i,j,t} + a_{t+1} \leq \{w_i z_t + R_t a_t\} \tau(a, z)$$

- ▶ Household in i with (a, z) : Indifferent between transition if wealth scaled by $\tau(a, z)$

Welfare Gains



Conclusion

- ▶ Introduce multi-country to Bewley economy
- ▶ Characterize contribution of incomplete markets and heterogeneity to trade elasticity
- ▶ Find gains of trade significantly higher relative to canonical trade models
 - ▶ Heterogeneity and risk contributing to higher trade elasticity for poor HHs pushes average gains up
- ▶ Log utility case looks like ACR result